

VA-ATACOM Supplementary Material: The DT-margin Heuristic Baseline

Anonymous submission

Table 1: DT-margin ablation on Goal (5 seeds).

Config	α	h	Mean $\bar{C}$	GO/5
Velocity-adaptive only	0.3	0	2.90	5/5
BRT-only	0.0	3	2.35	4/5
DT-margin (full)	0.3	3	0.96	5/5

The DT-margin Heuristic Baseline

VA-ATACOM’s velocity augmentation has a heuristic predecessor we call the **DT-margin** filter — the “DT-margin (heuristic)” row of Table 1 in the main paper. It inflates the keepout radius linearly with speed, $r_{\text{eff}}(\mathbf{v}) = r_{\text{base}}(1 + \alpha \|\mathbf{v}\|)$, projects out the inward radial component, and adds the same sim-BRT reward shaping. As shown in the main paper’s design study, this linear margin is the *first-order* approximation of VA-ATACOM’s quadratic braking constraint; under fair hazard-radius matching it reaches 13/15 GO — two Push seeds short of VA-ATACOM’s 15/15, illustrating that the linearisation almost captures the right behaviour but leaves a residual gap that only the exact quadratic constraint closes. We characterise it here for completeness — its two tunable knobs (α and the BRT horizon h) and their interaction with the control rate. These findings motivated, and are superseded by, the parameter-free velocity-augmented manifold of the main paper.

Ablation of the two components

Table 1 (Fig. 1) shows both components contribute: velocity-adaptive margin alone reaches 5/5 GO at higher variance (2.90); BRT alone 4/5 GO (one M1 collapse at $\bar{C}=7.98$); the full heuristic 0.96 cost / 100% GO.

Δt sensitivity

Table 2 corroborates the M1 chord-excursion premise of the main paper’s failure-mode analysis (and hence Prop. 4): continuous-time ATACOM achieves zero cost at $\Delta t \leq 0.02\text{s}$ but fails at $\Delta t \geq 0.05\text{s}$, the phase transition matching the M1 threshold $\Delta t \|u\| \gtrsim \sqrt{d_{\text{safe}}} \cdot r$. The DT-margin heuristic tracks the safe regime at $\Delta t \in \{0.10, 0.20\}\text{s}$ with $h=3$. At $\Delta t=0.05\text{s}$ the picture is nuanced: the horizon rule prescribes $h^*=4$, but a 5-seed sweep finds $h=3$ at 3/5 GO (bimodal), $h=4$ at 0/5, and $h=5$ at 3/5 (one seed at 37.05). We attribute

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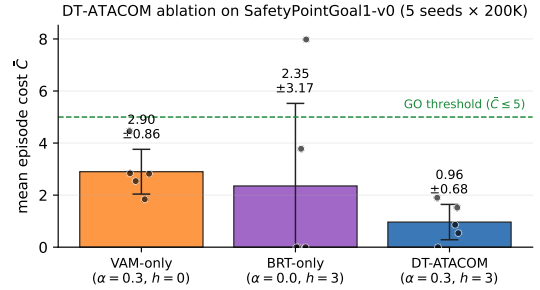


Figure 1: DT-margin ablation: both components are necessary. Mean episode cost $\bar{C}$ on SafetyPointGoal1-v0 (5 seeds $\times$ 200K env steps), bars are ± 1 std. VAM-only ($\alpha=0.3, h=0$) addresses M1 chord excursion only; BRT-only ($\alpha=0, h=3$) anticipates M2 drift but one seed collapses via M1; DT-margin ($\alpha=0.3, h=3$) combines both for lowest mean (0.96) and 5/5 GO.

Table 2: DT-margin cost vs. time step ($\bar{C} \pm \sigma$, 5 seeds, 50K steps).

Δt	ATACOM	DT-margin ($h=3$)	Note
0.01s	0.00	3.02	ATACOM OK
0.02s	0.00	0.00	ATACOM OK
0.05s	39.95	4.96 $\pm$ 6.08 (3/5 GO)	Basin regime
0.10s	13.68	2.60	DT-margin OK
0.20s	—	1.33 $\pm$ 1.50	$h \gg h^*$

this to a *BRT-shaping basin*: near the penetration horizon the lookahead reports unavoidable penetration along multiple directions every step, integrating spurious penalty into the policy gradient and trapping training.

α/h sensitivity

Sweeping α and h on Goal (5 seeds $\times$ 200K):

Only configurations at least one step above the minimum on *both* axes ($\alpha \geq 0.1, h \geq 2$) reach 5/5 GO; setting $\alpha=0$ drops to 4/5 with a seed at $\bar{C}=8.46$. The over-provisioned config ($\alpha=0.5, h=5$) *regresses* to 4/5 ($\bar{C}=9.26$ outlier) — large- h BRT shaping can integrate spurious penalty (see the Δt sensitivity above) and destabilise training. The DT-margin recipe

Table 3: DT-margin α/h sensitivity on Goal (5 seeds $\times$ 200K).

Config	α	h	mean $\bar{C}$	GO/5
no margin	0.0	2	3.37 ± 2.81	4/5
short horizon	0.05	1	1.94 ± 1.78	4/5
both at minimum	0.05	2	2.36 ± 2.60	4/5
above minimum	0.1	2	1.30 ± 1.66	5/5
α generous	0.3	1	1.72 ± 1.46	5/5
paper choice	0.3	3	0.96 ± 0.68	5/5
over-provisioned	0.5	5	3.00 ± 3.38	4/5

is thus robust only with margin on both knobs; VA-ATACOM eliminates α entirely, replacing it with the calibrated $a_{\max}$.